Analytics Methods and Frameworks Project

Overview

For this project, you will be developing a predictive framework and making strategic recommendations for either a struggling company, NovaBank, or a business scenario of your choosing. While you can fully hand code this project if you wish, you are highly encouraged to utilize leading AI tools (e.g., ChatGPT, Cursor, Claude) to accelerate development. You can complete this project either individually or as a group of no more than **four** people.

Learning Outcomes

When completed successfully, this project will enable you to:

- Frame business challenges as analytics problems, with clear KPIs and success measures.
- Prepare and explore data to ensure quality and uncover key patterns.
- Build and evaluate models (forecasting, classification, clustering, association rules, machine learning) to extract insights and highlight trade-offs.
- Translate results into decision rules that balance value, fairness, and explainability.
- Communicate findings effectively through visuals, an executive memo, and a presentation.

Project Description

Executives today face hard trade-offs in allocating scarce analytics budgets. Your analysis provides decision rules that balance financial impact, fairness, and explainability, ensuring leadership can act with confidence. This project is your opportunity to apply analytics methods to address real business problems with clarity and impact. You'll be

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evaluated on the quality of your solution, the soundness of your reasoning, and how well your insights support decision-making. *You can choose for the subject of your project either a business scenario and dataset of your choosing, or the NovaBank business case, detailed below.*

NovaBank: Predictive Retention at Scale
NovaBank's margins are under pressure, and customer churn continues to erode profitability. Executives want to shift from reactive retention to proactive intervention, spotting at-risk customers early enough to act. You are asked to develop a predictive framework that goes beyond "which model performs best." The goal is to create a system that not only identifies churn risk but also informs real-world decisions across operations, and customer success.

Your work must deliver:

- A clear recommendation on which customers to prioritize for retention efforts and how confident executives should be in acting on those signals.
- A trade-off analysis of false positives (wasted offers) vs. false negatives (lost customers).
- An explanation framework that makes results understandable and defensible for non-technical stakeholders.

The dataset for this business case can be found here. If you feel the provided dataset is not sufficient for certain methods or insights, you may simulate additional data (e.g., adjust churn rates, add synthetic features, or create scenario variations) as long as you document your assumptions.

NovaBank is a fictional company, and you are not expected to apply specific banking regulations. Instead, assume that any recommendation must be transparent, fair, and defensible if questioned by compliance officers or other non-technical stakeholders.

Assignment Requirements

You will be submitting a single PDF, consisting of the following deliverables:

- A one-page executive memo outlining your decision, rationale, impact, and risks
- A link to a slide appendix of between 8-10 slides that details your methods, evidence, scenarios, business implications, and AI usage.
- A link to your reproducible notebook (GitHub or Google Drive)

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Resources

- Before you begin, you may want to brush up on Python—Python courses can be accessed by searching "Python" in the Courses tab on your dashboard. You may also want to take Learn to Code with ChatGPT and Exploring the Data Science Landscape.
- Watch the webinar session recording below for additional project guidance: https://us02web.zoom.us/rec/share/F1t3BuAEDqpkw3LlhJXYLGIDReAbR8JffhQoszs077alTQu_3lUCT3xxkEbGJudJ.7vRyaajxBZP0FmW4?startTime=1758582786000 (Passcode: !RVR!fa3)
    - Slide Deck

Step-by-Step Guidance

1. Problem Framing and Scope
    - Reframe the problem, and write a short problem brief (about 5 sentences): context, decision to be made, who cares, timeframe, and constraints.
    - Translate the business problem into an analytics problem
    - Choose the right methods, and accordingly pick 1–2 success measures (e.g., forecast error, model accuracy, precision at top group, cluster quality, or rule strength).
    - Draft a specific action or policy you want your analysis to inform (e.g., "target top 20% at risk," "increase stock 15%").
    - Additional Resources:
        - "To Solve a Tough Problem, Reframe It" (HBR)
        - Chapter 2 ("Business Problems and Data Science Solutions") of Data Science for Business by Foster Provost & Tom Fawcett.
        - "AUC vs Precision-Recall Curve" - AI Competence
2. Data Readiness
    - Load and profile your dataset.
    - Define your target (e.g., churn flag, monthly sales, RFM table).
    - Create a simple train/test split
    - Apply light cleaning if needed (deduplicate, remove invalids, handle returns).
    - Write a short data dictionary (1–2 lines per field).
    - You're free to use R or Python, and work in Google Colab, Jupyter Notebook, RStudio, VS Code, or any IDE you prefer. You can also use AI copilots (like ChatGPT, Cursor, or Claude) to support your work, as long as you document how you used them.

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    - The Quantic courses Introduction to Machine Learning and Data Pre-Processing may also be helpful for this step.
3. Model Building: Baseline & Improved Models
    - Build simple baselines: for example, seasonal naïve or smoothing, logistic regression, RFM, k-means, or top association rules.
    - Record results in a short summary table.
      Develop one improved model (for example, ARIMA, tree/ensemble, refined clustering, or pruned rules).
    - Compare against the baseline and explain improvements in plain language.
    - (Optional) Add engineered features, calibration, or alternate methods for stretch. Deliver a results table (baseline vs. improved) and one diagnostic slide.
    - Additional resources:
        - fast.ai "Practical Deep Learning for Coders" course
4. Decision Framework
    - Pick a threshold or cutoff and link it to a clear business action.
    - Run a simple scenario test (e.g., demand shock, promo uplift, class imbalance) and note what holds vs. what shifts.
    - Translate results into practical decision rules (who, what, when, how much).
    - Draft a brief pilot plan with scope, KPI, and timeline.
    - Flag 2–3 key risks or assumptions executives should know.
    - Additional resources:
        - "Robustness and Sensitivity Analysis" (Chapter 13 in The Elements of Statistical Learning by Hastie, Tibshirani, Friedman)
5. Quality Check and Delivery
    - Use a proper train/test split.
    - Show baseline + improved model.
    - Tie metrics back to a clear decision rule.
    - Keep notebook reproducible, with AI usage logged.
    - Review deliverable requirements in the "Assignment Requirements" above before submitting
    - The Quantic course Introduction to Git and Github is valuable for this step
    - Additional resources:
        - 500+ Projects with Code

Frequently Asked Questions

Can we extend the project further for our capstone?
Yes. You are allowed to build on this project for your capstone if you find the business problem meaningful.

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How should we comment and share our interpretations?
Add clear annotations and comments directly in your code. Use text cells or notes to summarize interpretations and neatly answer each guiding question.

What if the dataset feels too limited?
You may simulate additional data (e.g., new features, adjusted distributions, scenario variations) as long as you document your assumptions and explain the purpose.

How do we document AI or copilot use?
Keep a short log of prompts or tasks where AI assisted you, and include it either in your slides (as a closing section) or as notes in your notebook. Focus on what AI contributed and what you learned.

Submission Guidelines

Your submission should consist of a single PDF. To submit your project, please click on the "Submit Project" button on your dashboard and follow the steps provided. If you are submitting your project as a group, **please ensure only ONE member submits on behalf of the group**. Please reach out to msba+projects@quantic.edu if you have any questions. Project grading typically takes about 3-4 weeks to complete after the submission due date. There is no score penalty for projects submitted after the due date, however grading may be delayed.

Plagiarism Policy

Here at Quantic, we believe that learning is best accomplished by “doing”—this ethos underpinned the design of our active learning platform, and it likewise informs our approach to the completion of projects and presentations for our degree programs. We expect that all of our graduates will be able to deploy the concepts and skills they’ve learned over the course of their degree, whether in the workplace or in pursuit of personal goals, and so it is in our students’ best interest that these assignments be completed solely through their own efforts with academic integrity.

Quantic takes academic integrity very seriously—we define plagiarism as: “Knowingly representing the work of others as one’s own, engaging in any acts of plagiarism, or referencing the works of others without appropriate citation.” This includes both misusing or not using proper citations for the works referenced, and submitting someone else’s work as your own. Quantic monitors all submissions for instances of plagiarism and all plagiarism, even unintentional, is considered a conduct violation. If you’re still not sure about what constitutes plagiarism, check out this two-minute presentation by our

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librarian, Kristina. It is important to be conscientious when citing your sources. When in doubt, **cite**! Kristina outlines the basics of best citation practices in this one-minute video. You can also find more about our plagiarism policy here.

Project Rubric

Scores 2 and above are considered passing. Students who receive a 1 or 0 will not get credit for the assignment and must revise and resubmit to receive a passing grade.

Score	Description
5	• Provides a highly persuasive and well-supported analytics case. • Clear strategic alignment with the defined business challenge and KPIs. • Strong executive appeal with a compelling narrative and actionable recommendations. • Results are grounded in sound baselines, improved models, and sensitivity/simulation analysis. • Clear explanation of trade-offs, risks, and implications for decision-making. • Submission includes a polished 1-page memo, well-structured slide appendix (≤10), and a reproducible notebook. • Demonstrates excellent application of course concepts (predictive modeling, segmentation, forecasting, or association rules) with depth and insight. • Professional communication: concise, confident, and executive-ready.
4	• Provides a solid and reasoned analytics case. • Good alignment with the business problem and KPIs. • Narrative and recommendations are clear, though may lack some persuasiveness or depth. • Results include baselines and at least one improved model; sensitivity or scenario analysis present but less thorough. • Trade-offs and risks are discussed but may be less fully developed. • Submission is complete: memo, slides, notebook; minor polish issues. • Applies course concepts well, though some methods or interpretations could go deeper. • Communication is clear, with minor lapses in conciseness or executive focus.

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3	• Provides a reasonable but somewhat basic analytics case. • Partial alignment with business problem; KPIs may be underdeveloped. • Narrative is understandable but lacks strong executive appeal or actionability. • Results include baselines; improved models may be limited or less justified. • Sensitivity/simulation is minimal or missing. • Submission meets core requirements (memo + slides + notebook) but lacks refinement. • Applies course concepts in a basic or incomplete way. • Communication is serviceable but less polished or engaging.
2	• Analytics case is incomplete or weakly reasoned. • Limited or unclear connection to business needs/KPIs. • Narrative lacks clarity and recommendations are vague or unsupported. • Results may include one model but no baselines, or unclear metrics. • No evidence of sensitivity or trade-off analysis. • Submission is incomplete or poorly structured. • Minimal application of course concepts; key gaps in understanding. • Communication is unclear, overly technical, or confusing for executives.
1	• Analytics case is irrelevant, unsupported, or incoherent. • No meaningful alignment with business goals. • Narrative is disjointed or missing; no actionable insights provided. • Models or results are absent or unusable. • Fails to apply course concepts or shows major misunderstandings. • Submission missing major deliverables. • Communication is very difficult to follow.
0	• The student either did not complete the assignment, plagiarized all or part of the assignment, or completely failed to address the project requirements.

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